

ds_final_data_cleaning

Chuyuan XU

2025-12-02

import the county and state data

```
state_r = read_sf(str_c(import, "/tl_2020_us_state")) |>
  janitor::clean_names() |>
  as.data.frame() |>
  filter(stusps == "NY") |>
  select(statefp, stusps, name)

stsfy_ny = state_r |>
  distinct(statefp) |>
  pull()

county_ny = read_sf(str_c(import, "/tl_2024_us_county")) |>
  janitor::clean_names() |>
  filter(statefp %in% stsfy_ny) |>
  select(statefp, countyfp, fips = geoid, name, namelsad)

fips_ny = county_ny |>
  distinct(fips) |>
  pull()
```

import all NY meteor data and filter county in NY state

```
meteor_path = str_c(import, "/meteorology")
meteor_files = list.files(meteor_path)

meteor_16to20 = read_parquet(str_c(meteor_path, '/', meteor_files[1])) |>
  janitor::clean_names()

for (i in 2:length(meteor_files)){
  meteor_temp = read_parquet(str_c(meteor_path, '/', meteor_files[i])) |>
    janitor::clean_names()

  meteor_16to20 = bind_rows(meteor_temp, meteor_16to20)
}

meteor_16to20 = meteor_16to20 |>
  filter(county %in% fips_ny) |>
  mutate(
    fips = county
  ) |>
  select(county, fips, date, everything()) |>
```

```

arrange(county, fips, date)

# still too large (more than 50mb)
meteor_16to20_sf = meteor_16to20 |>
  left_join(county_ny, by = "fips") |>
  st_as_sf()

```

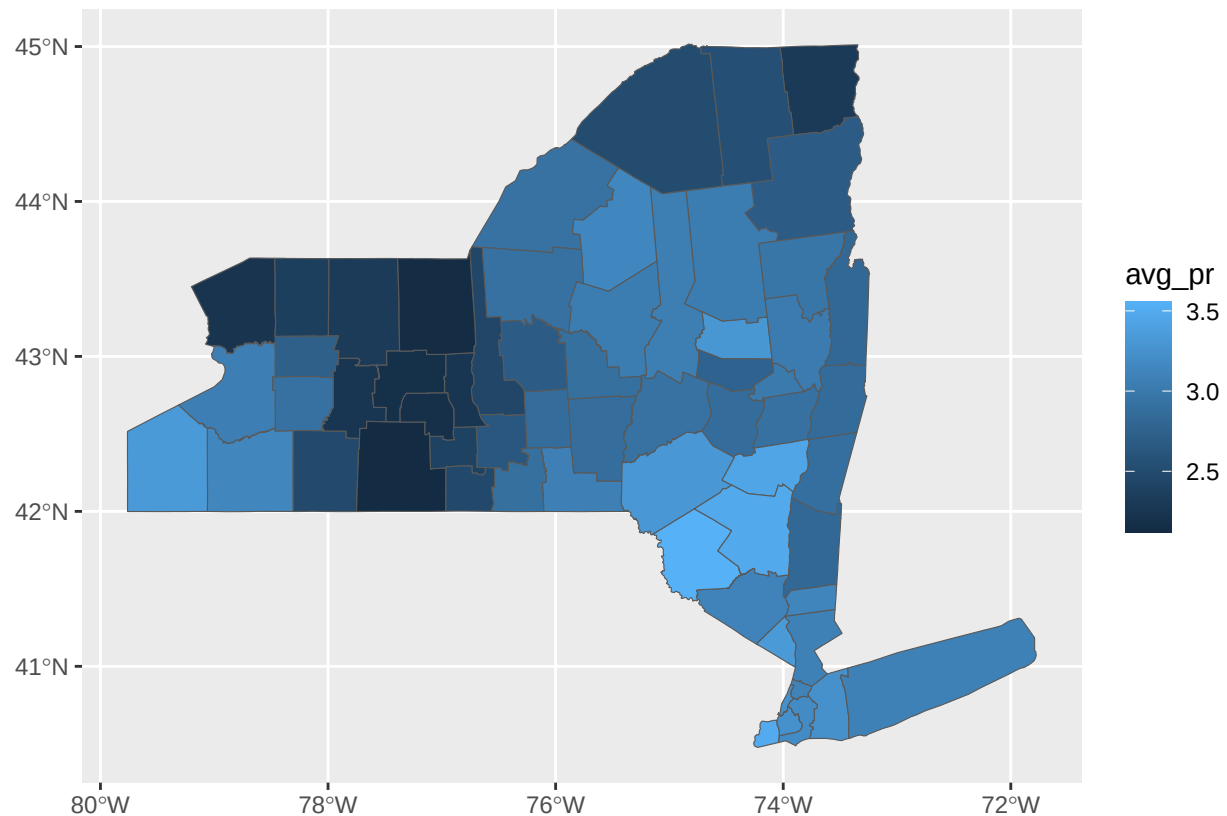
try some visualization to test the dataset

```

meteor_16to20_sf |>
  filter(year(date) == 2020) |>
  group_by(fips) |>
  summarise(
    avg_pr = mean(pr)
  ) |>

ggplot() +
  geom_sf(aes(fill = avg_pr))

```



I think it works?

output tht data

```

dsn_meteor = str_c(output, "/meteor_2016to20_NY state.csv")

write_csv(
  meteor_16to20,

```

```

    dsn_meteor,
    na = "NA",
    append = FALSE
)

dsn_geo = str_c(output, "/NY county_fips")

st_write(
  county_ny,
  dsn_geo,
  driver = "ESRI Shapefile",
  append = FALSE
)

```

Check

```

file_check = read_csv(
  '/Users/chuyuanxu/Desktop/Columbia/25_fall/BIST8105_DS I/final project/meteor_2016to20_NY state.csv',

## Rows: 20 Columns: 13
## -- Column specification -----
## Delimiter: ","
## dbl  (12): county, fips, sph, vpd, tmmn, tmmx, pr, rmin, rmax, srads, vs, th
## dtm   (1): date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

```